
Building Trustworthy Models

PRACTICAL STATISTICS EVERY DEVELOPER SHOULD KNOW

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- **Master's of Statistical Science**
- **10 years experience in data science and machine learning**
 - **Building products to support stakeholder and business strategy**
 - **Intersection between technical and non-technical teams**



WHERE WE'RE GOING

- **Basic statistical building blocks**
- **Model fitting process**
- **How do we know what is “best”?**

MODEL FITTING

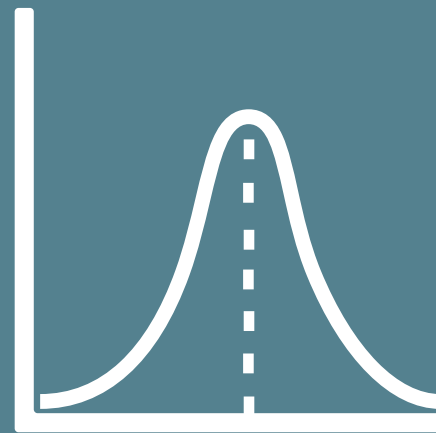
Mathematical description of the relationship between inputs and an outcome that is built from data. Approximation that captures a pattern but will never be perfect.



Data
Collection



Exploratory
Data Analysis



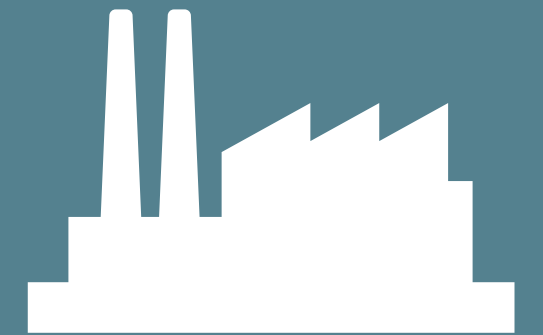
Data
Preparation



Model
Training

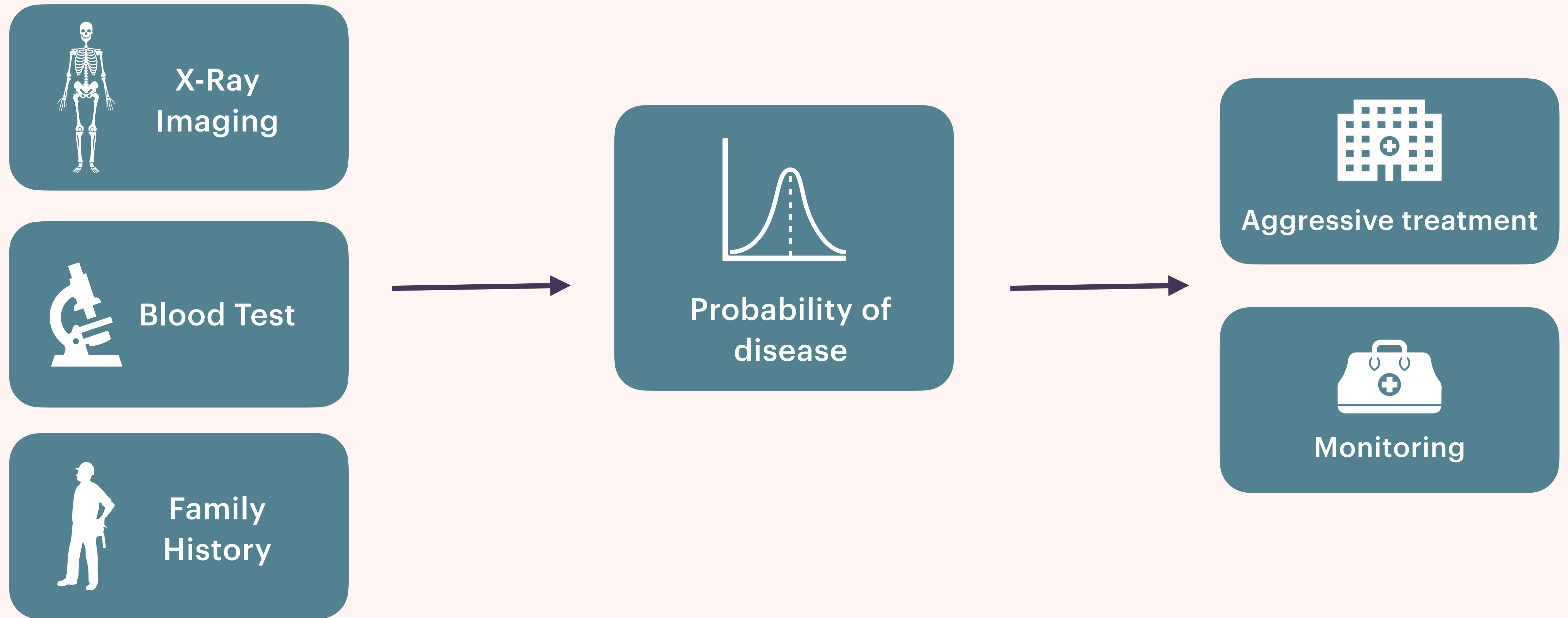


Model
Evaluation

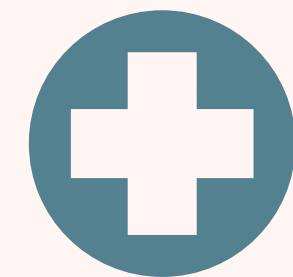


Ship to
Production

PREDICTING DISEASE



WHICH MODEL SHOULD YOU PICK?



Predicting Disease

5% of population has disease

Model A

Overall Accuracy: 95%

Always predict negative
500 true positives, 10k true
negatives

Model B

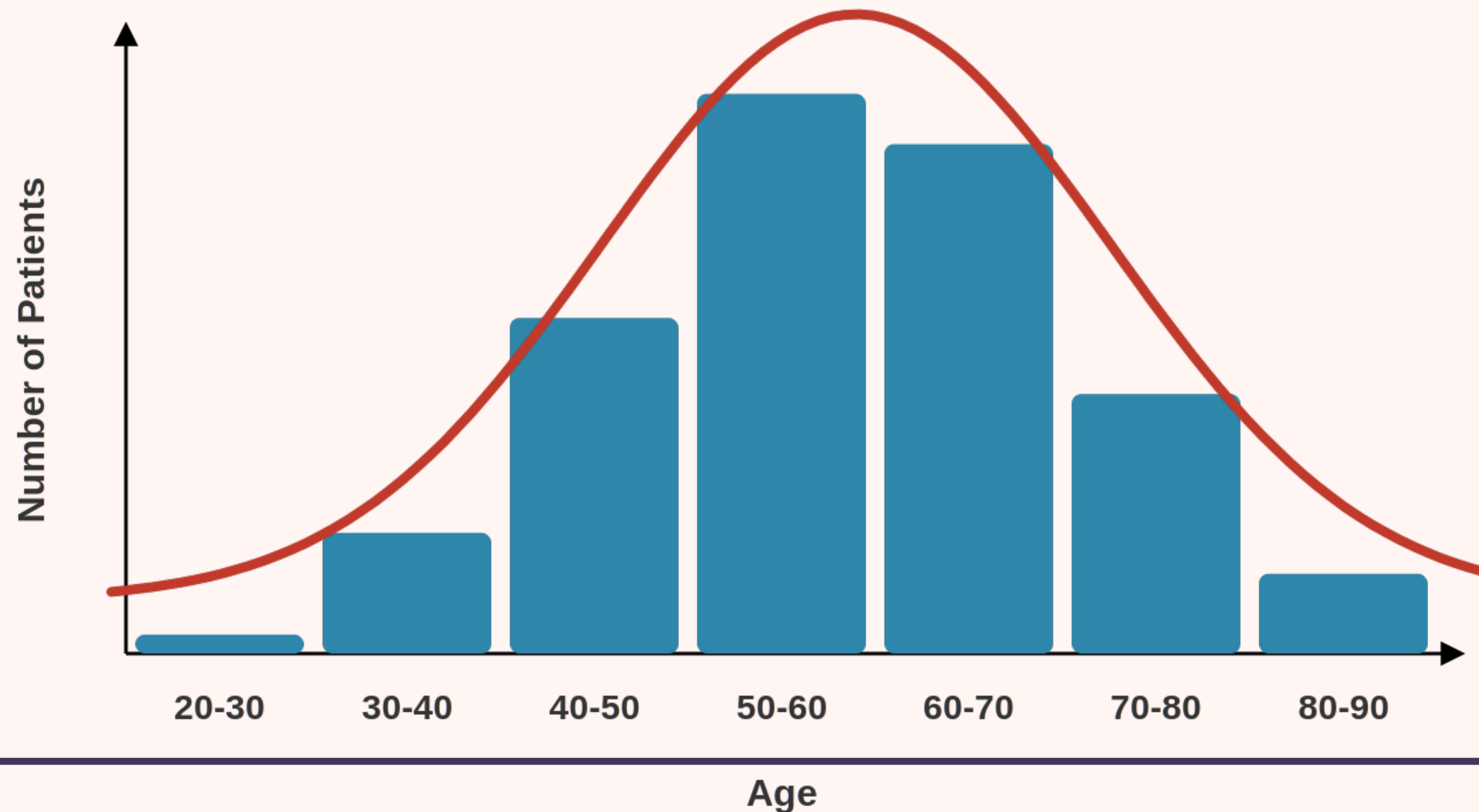
Overall Accuracy: 72.5%

Actually identifies
some true positives

DISTRIBUTION

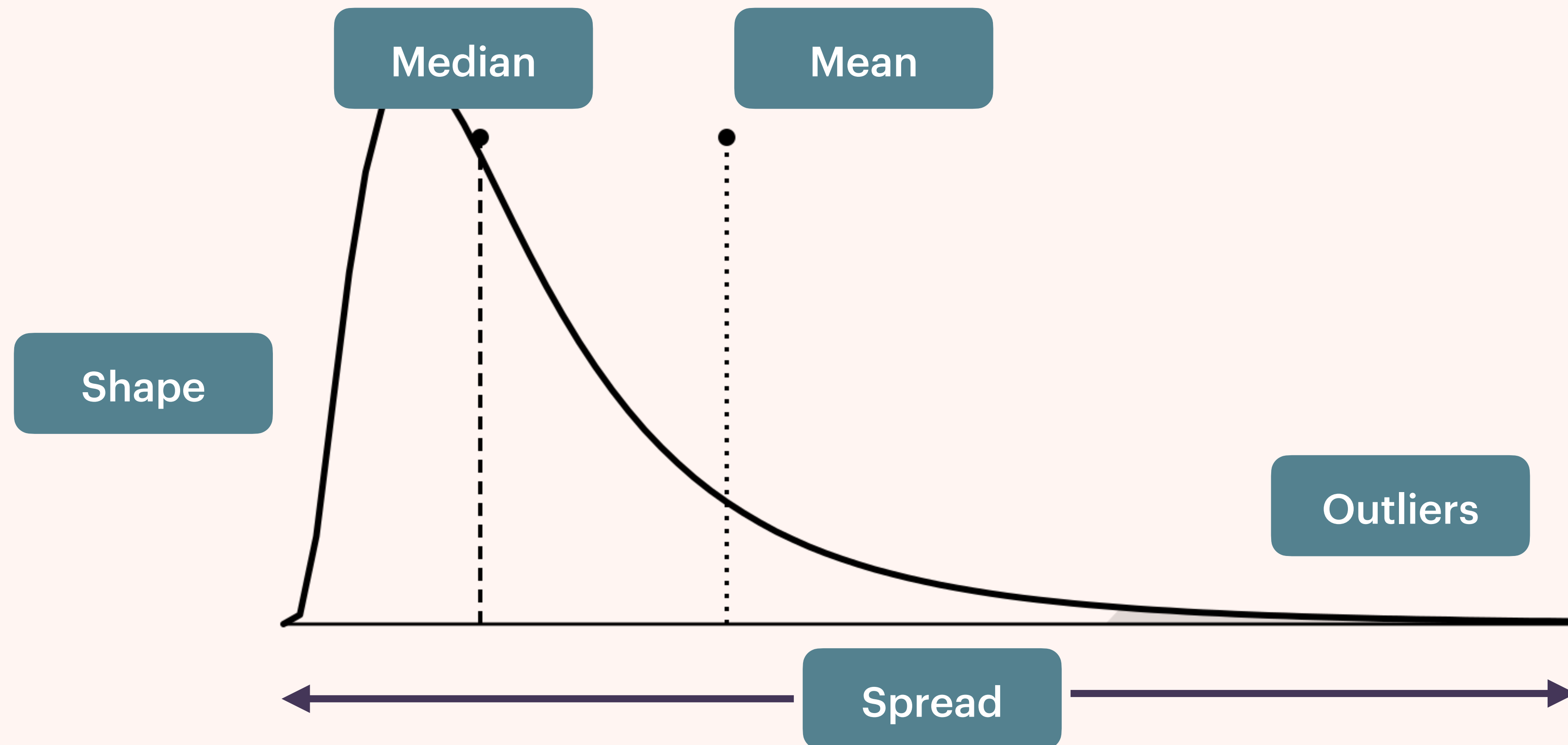
DISTRIBUTION

What values you see and how often they appear



DISTRIBUTION

What does it tell us?



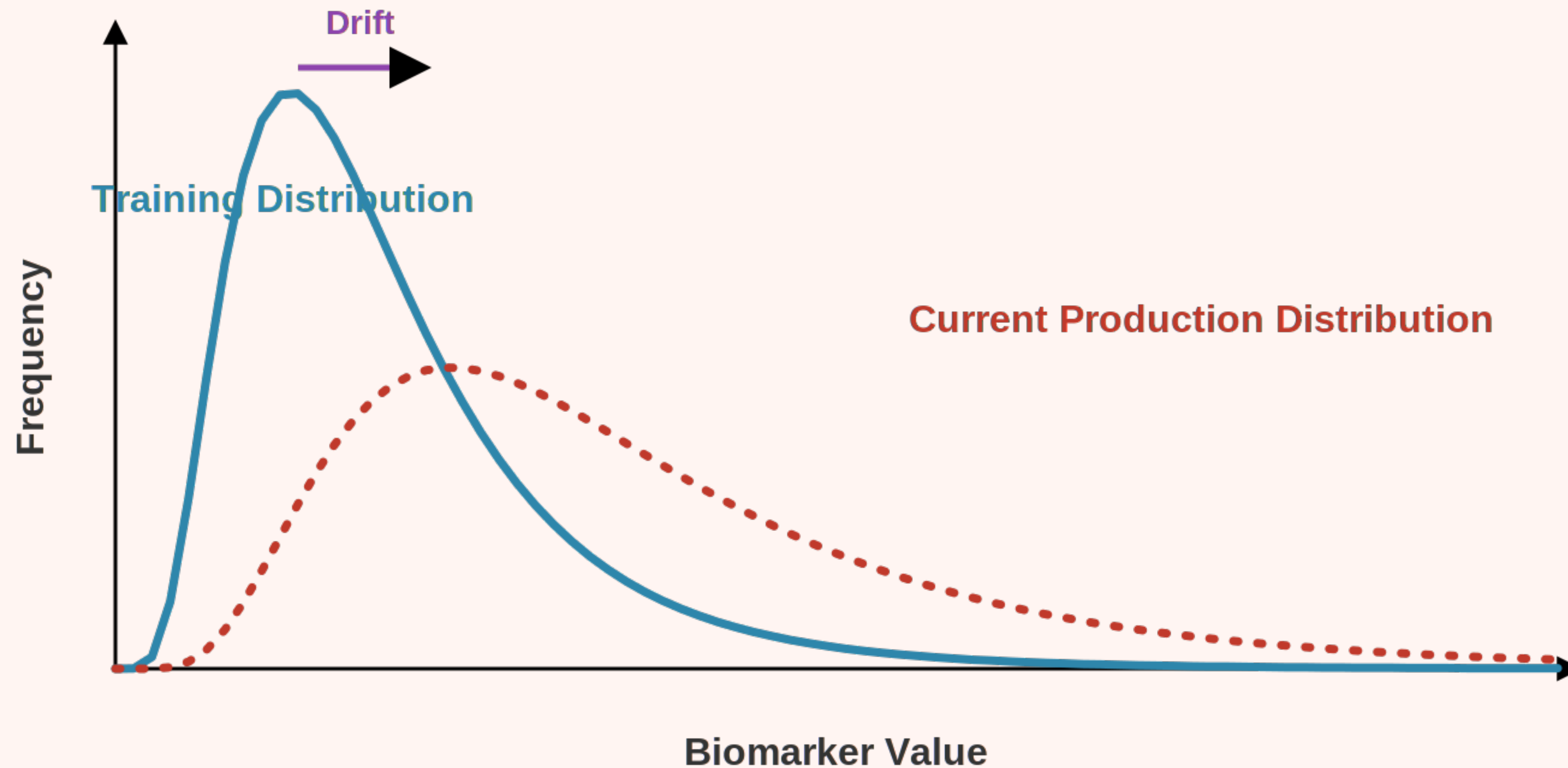
DISTRIBUTION

Why is it important?

- What does “typical” data look like?
- What counts as an outlier or noise?
- Has anything changed since model training?

DISTRIBUTION

Why is it important?



BIAS/VARIANCE TRADEOFF

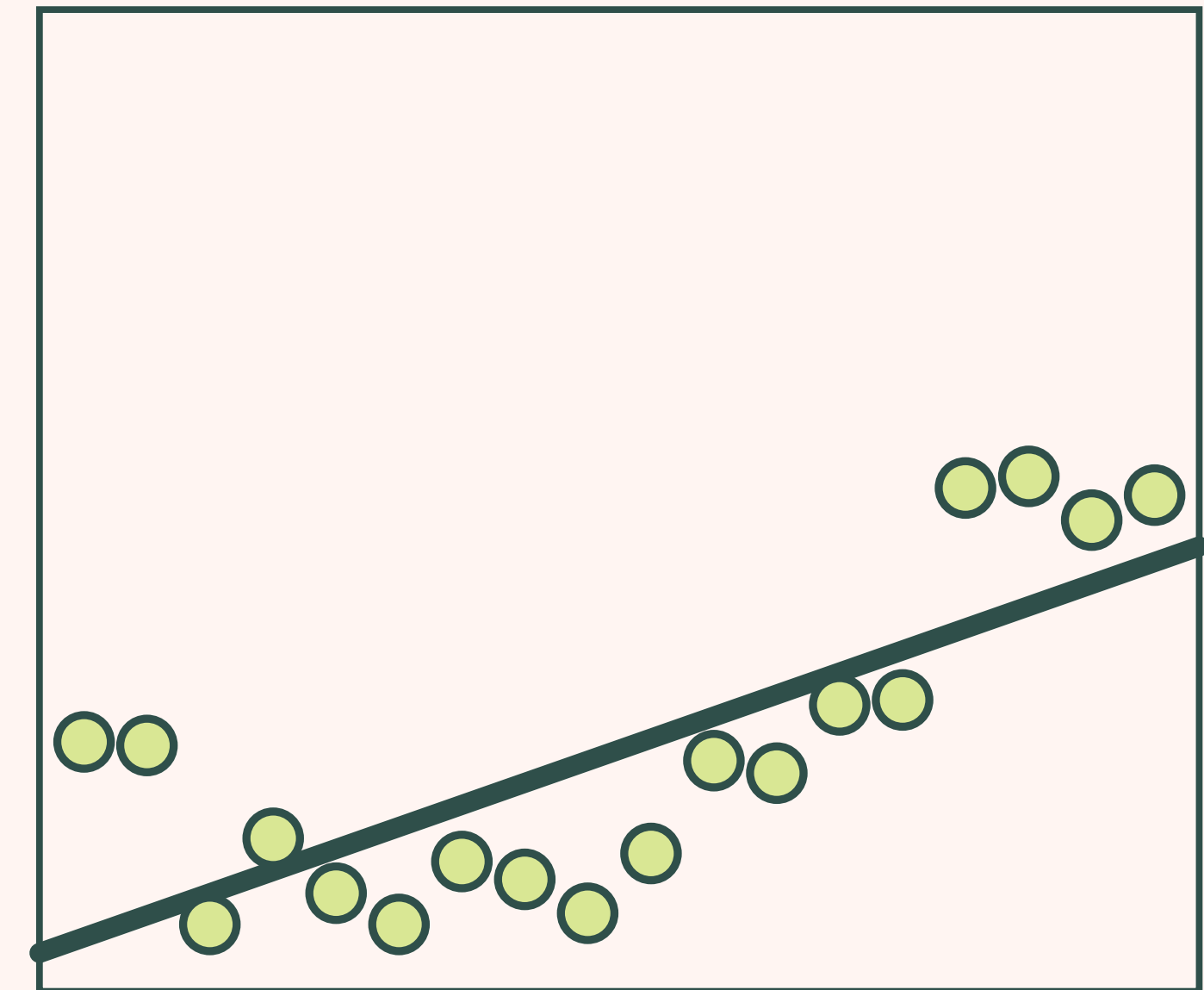
BIAS

How far off model prediction is from truth

$$\text{Bias} = E[\hat{f}(x)] - f(x)$$

Model's predictions on
average

Real underlying
relationship

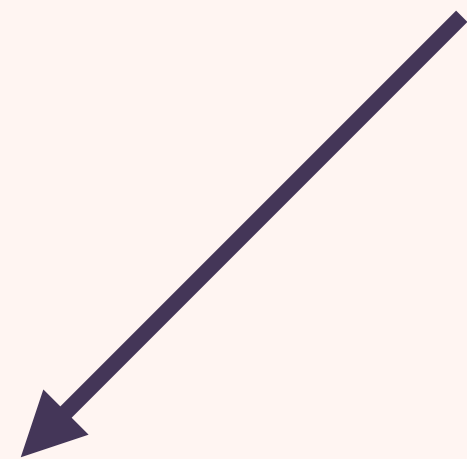


High Bias

VARIANCE

What change would you see if model trained on different data

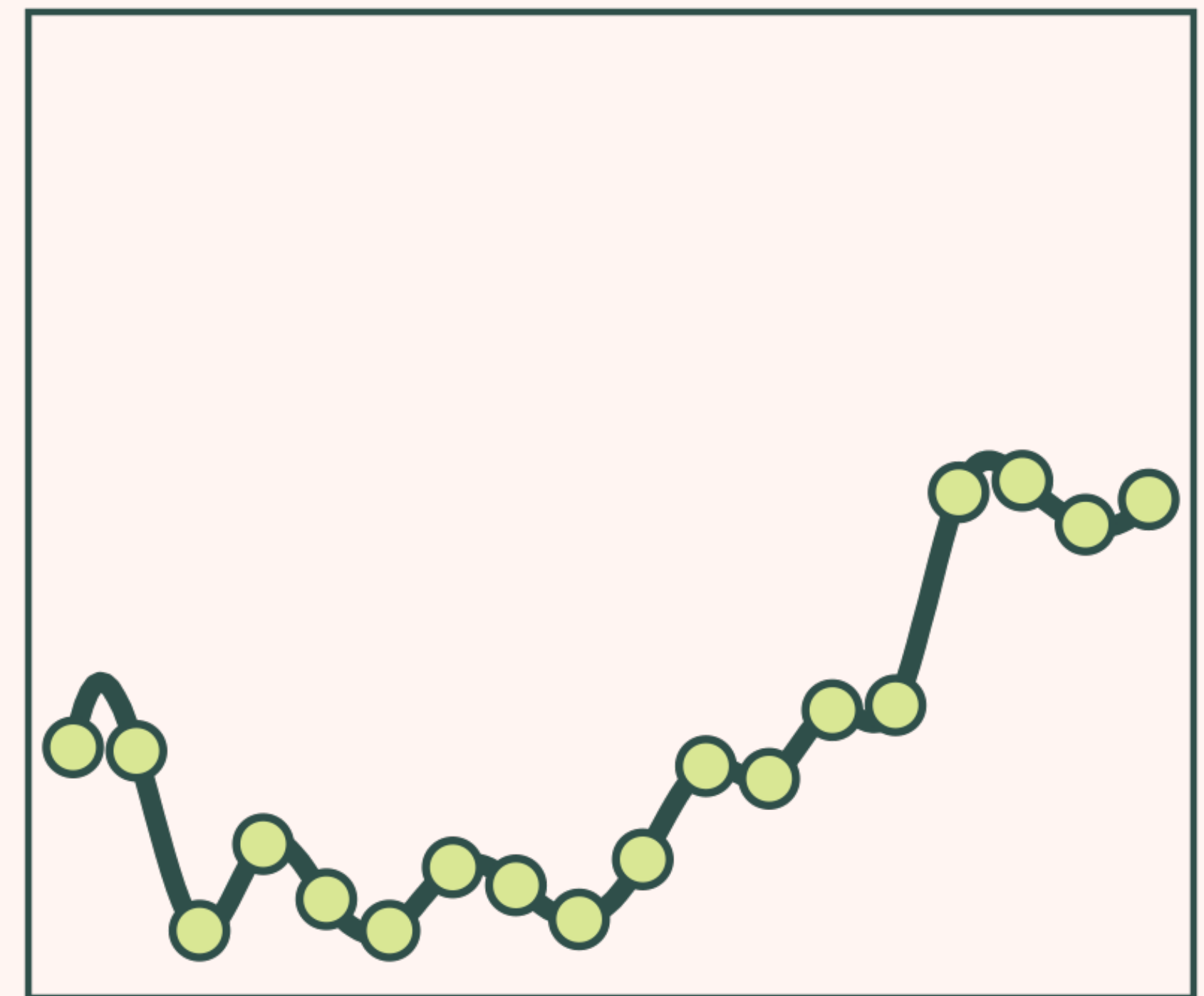
$$\text{Variance} = E[(\hat{f}(x) - E[\hat{f}(x)])^2]$$



Model prediction



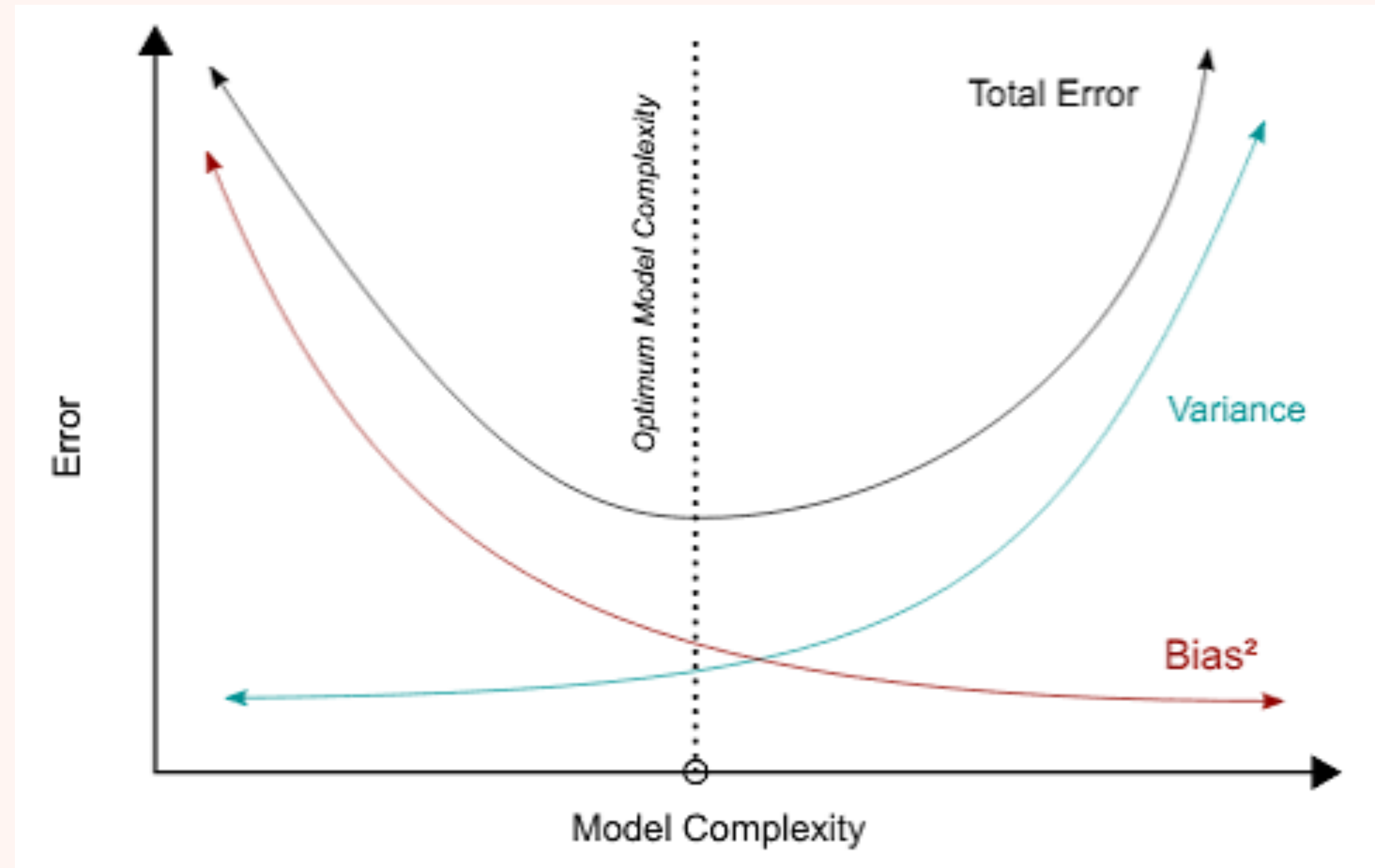
Model's predictions on
average



High Variance

BIAS/VARIANCE TRADEOFF

$$\text{Error} = \text{Bias}^2 + \text{Variance} + \text{Irreducible Error}$$



BIAS/VARIANCE

What does it look like?

High Bias

- Missing patterns
- Model too simple
- Poor performance

High Variance

- Captures noise
 - Sensitive model
 - Bad with test data
-

BIAS/VARIANCE TRADEOFF

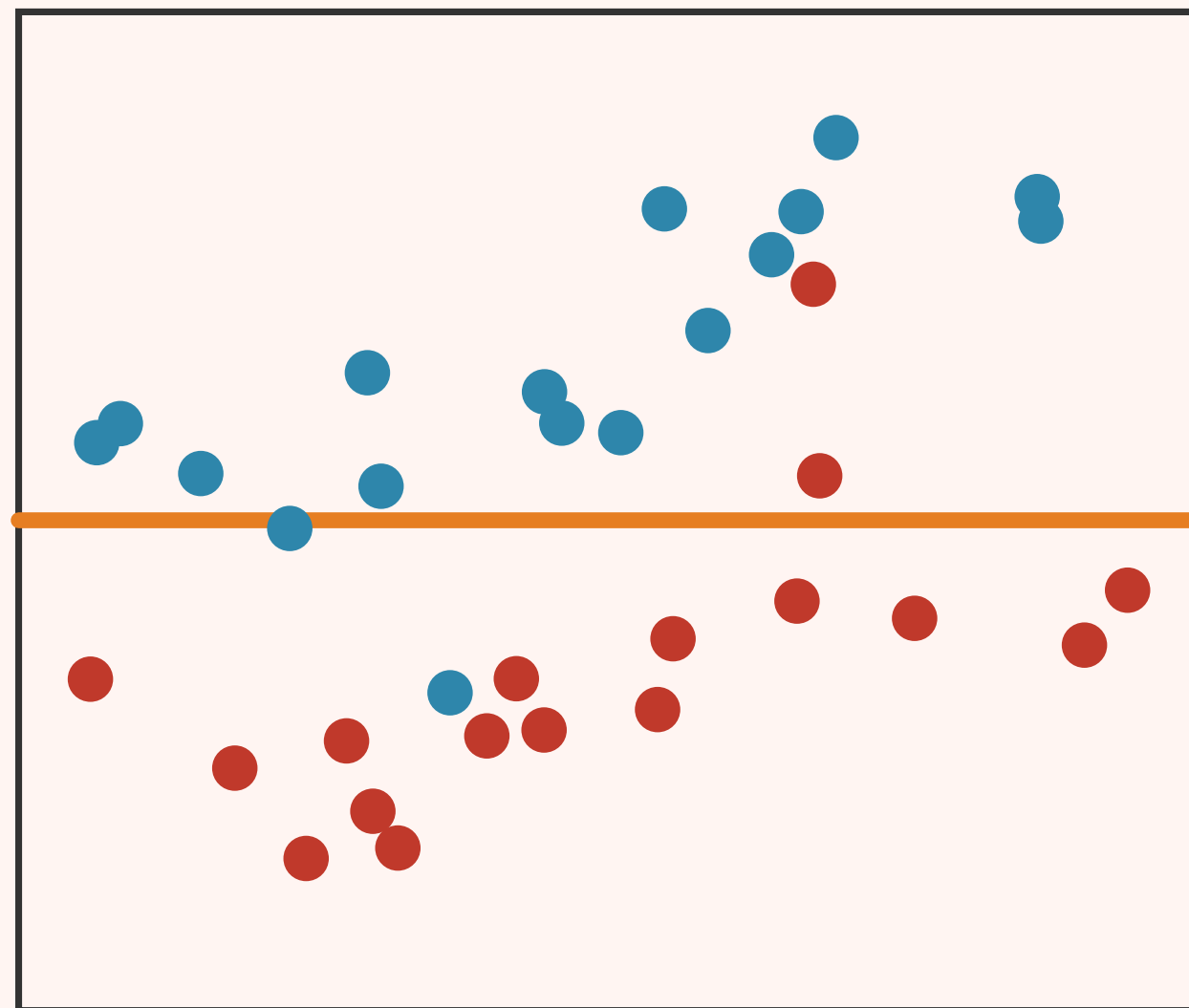
Why is it important?

Problem	Description	Use Case
High Bias	Model too simple	Predicts average risk for everyone
High Variance	Model too sensitive	Swings wildly with small differences in patients

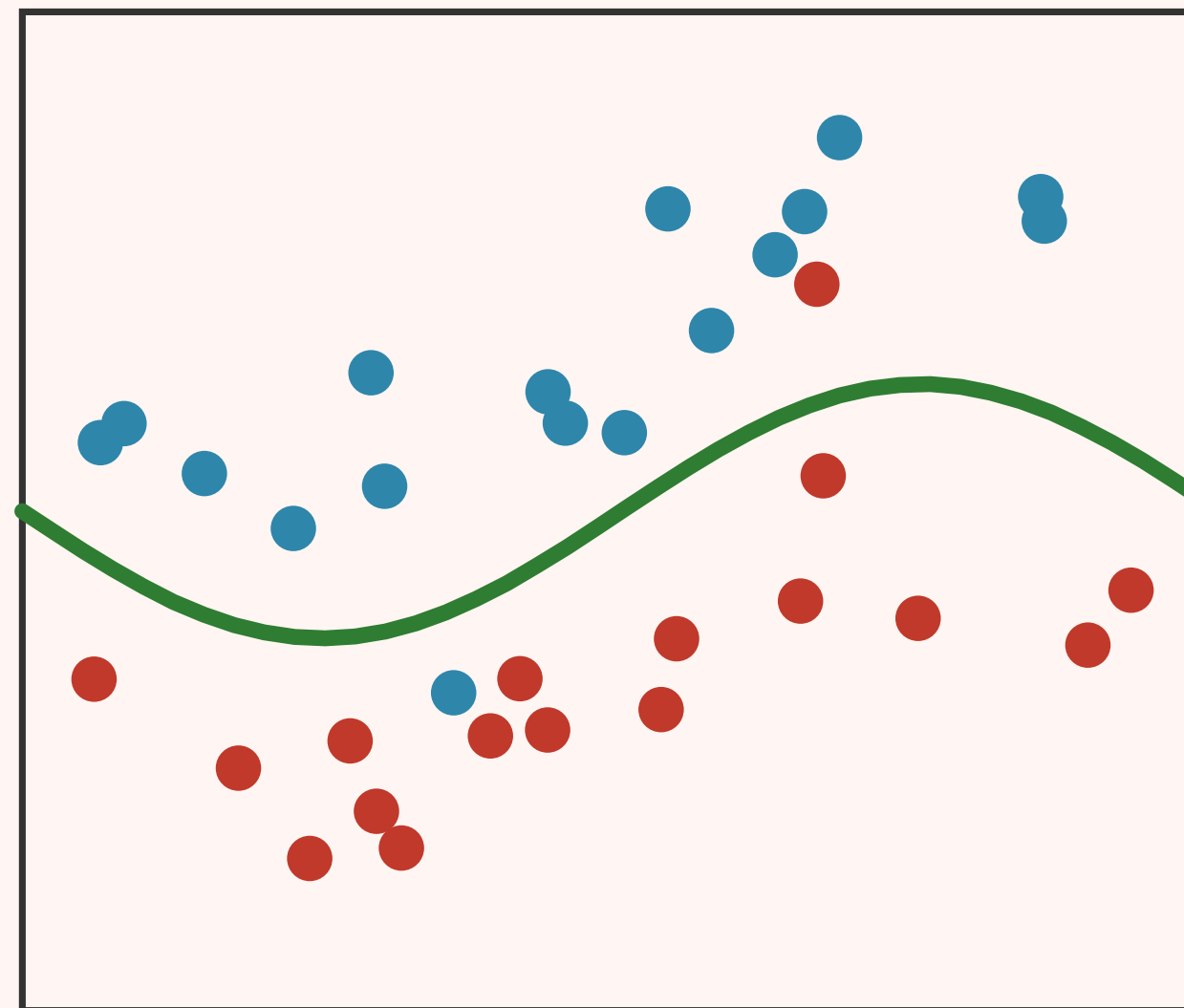
OVERFITTING

OVERFITTING

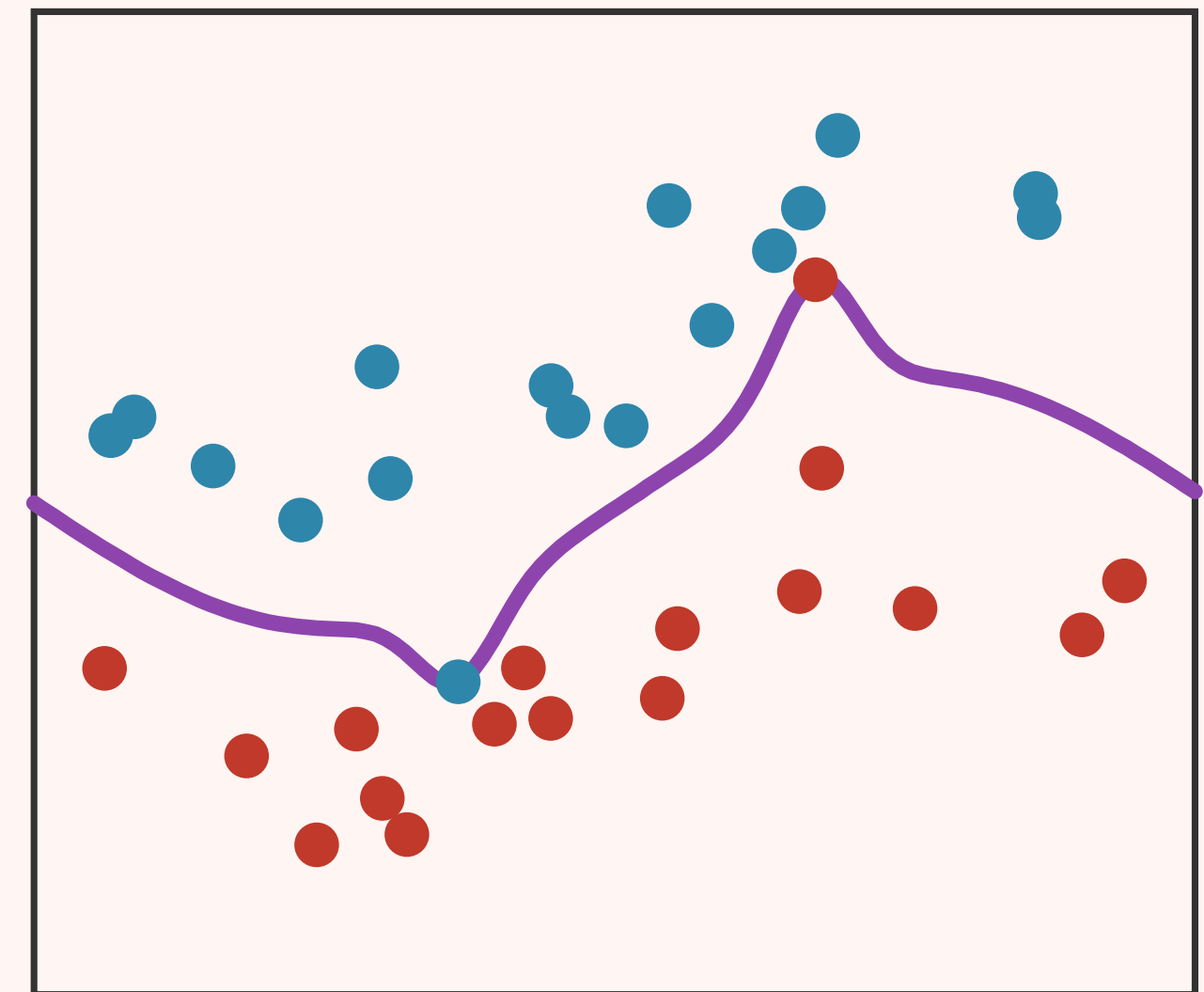
Model follows data too closely



Underfitting



Good Fit



Overfitting

OVERFITTING

Model follows data too closely

- High training accuracy that doesn't translate
- Model doesn't generalize well to new patients

OVERFITTING

How can we detect it?

Cross-validation

- Split data into folds
- Each fold takes a turn as validation
- Train on the rest, test on that fold
- Average the results

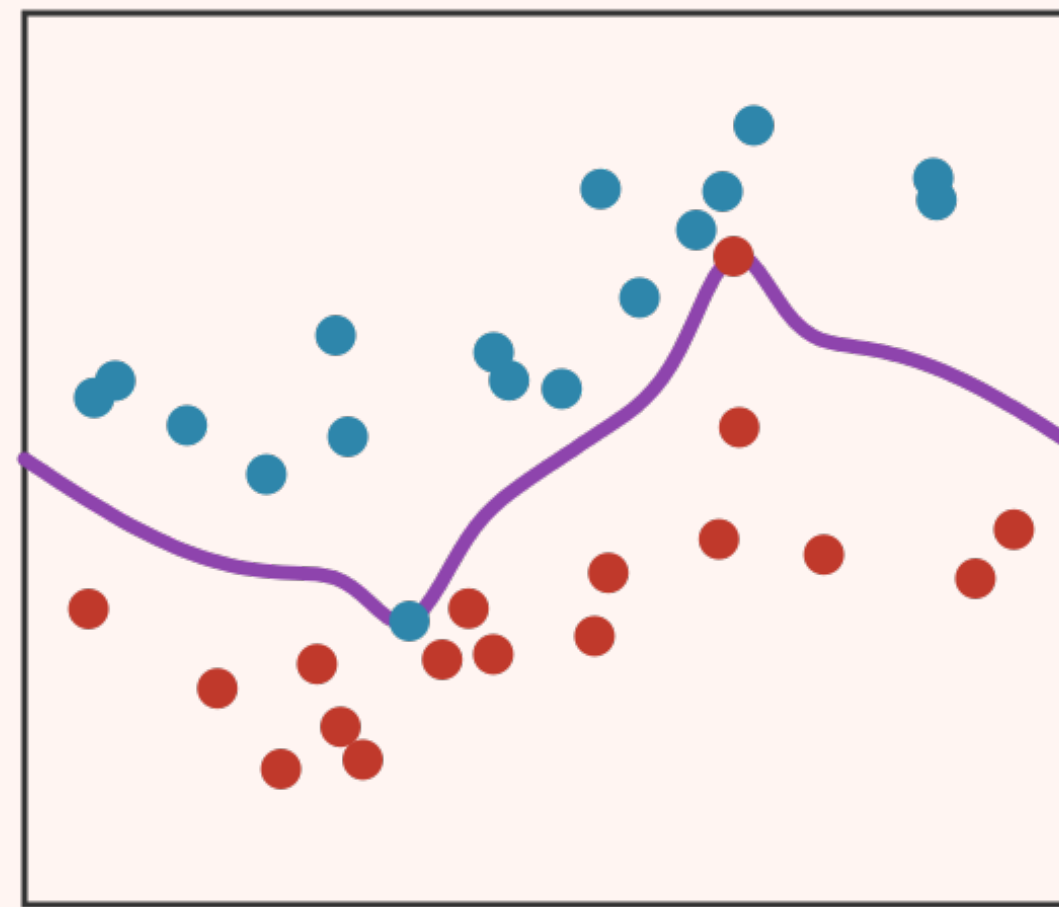


OVERFITTING

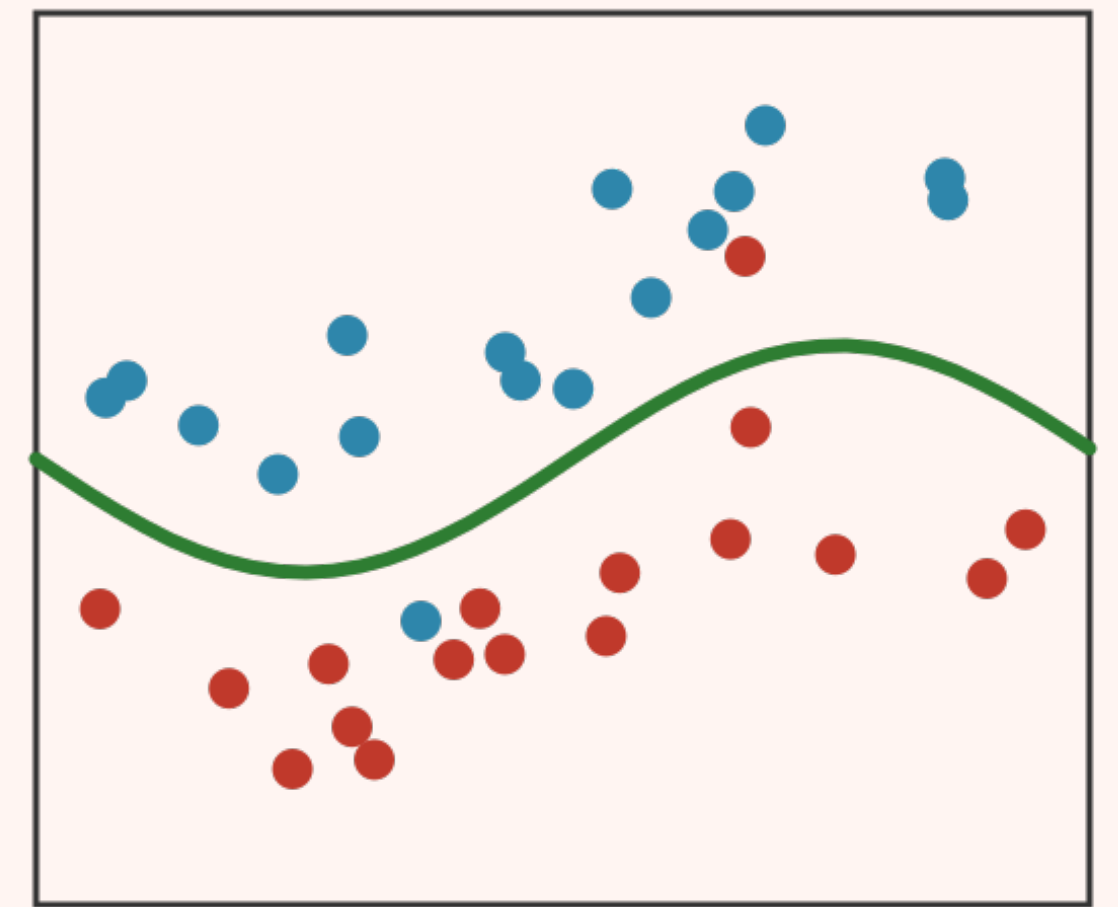
What can we do about it?

Regularization

- Penalize complexity
- Trade a little bias for a lot less variance



No Regularization



With Regularization

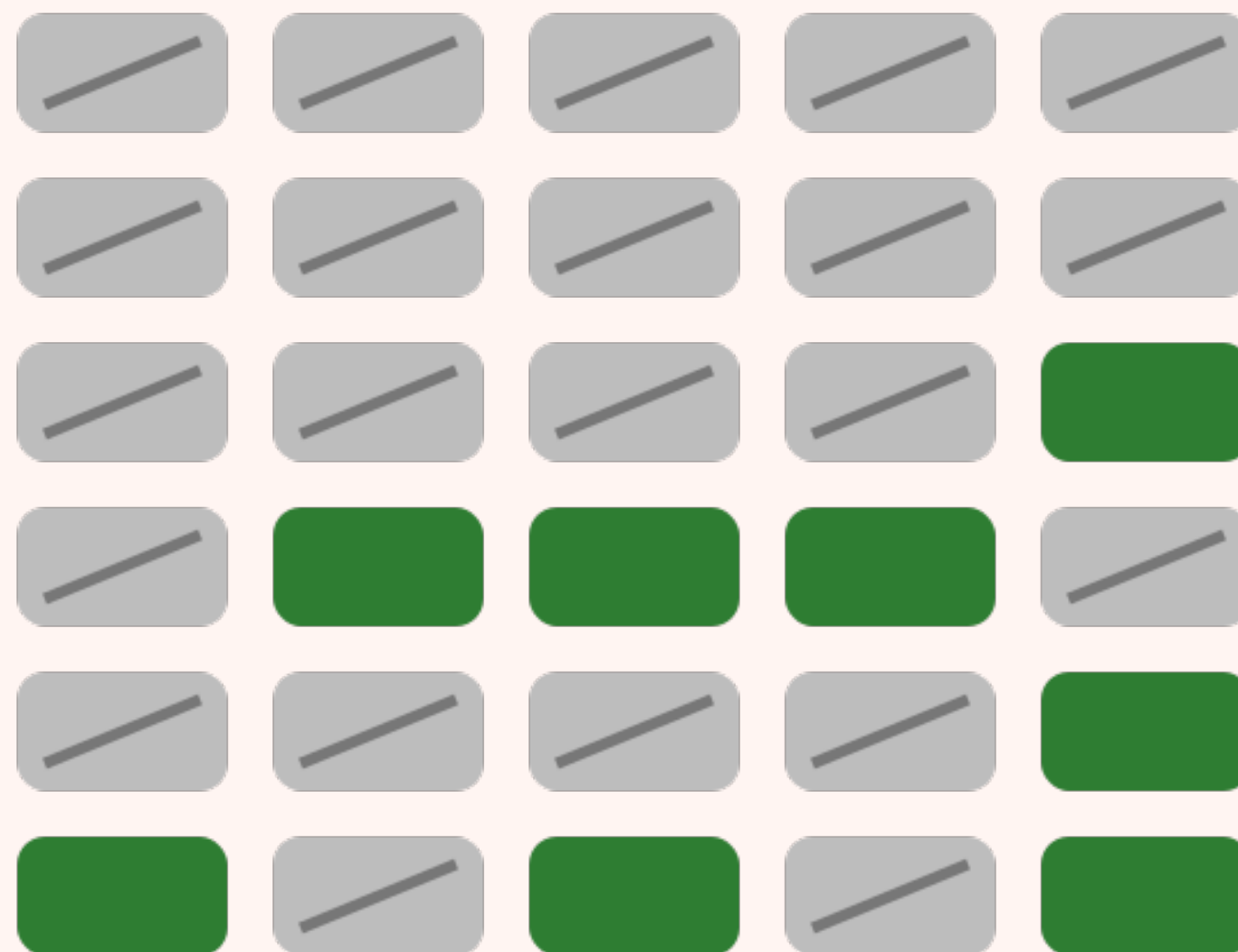
OVERFITTING

Regularization

Method	How It Works	Common Models
Mathematical Constraints	Adds penalties that force unimportant weights to zero	Logistic Regression, XGBoost
Structural Constraints	Restricts how complex or deep the model can become	Decision Tree, Random Forest
Training Restrictions	Randomly turns off components, stops process early	Neural Networks, Transformers

OVERFITTING

30 Biomarkers



 Kept  Dropped to zero

Regularization



8 Remain



EVALUATION METRICS

EVALUATION METRICS

Is this model “good”?

Confusion Matrix

		<i>Predicted</i>	
		Disease	Healthy
<i>Actual</i>	Disease	9	1
	Healthy	274	716

Error Types

Type 1: False Positive
No disease but predicts disease

Type 2: False Negative
Disease but predicts no disease

EVALUATION METRICS

Is this model “good”?

When a model predicts a positive outcome, how often is it actually right?

Precision

$$\frac{TP}{TP+FP}$$

A: 2.4%

B: 10%

If you are a positive, how likely is the model to predict that?

Recall

$$\frac{TP}{TP+FN}$$

A: 3.1%

B: 90%

EVALUATION METRICS

Metric	Description	Use Case
Precision	How many flagged patients were sick?	False alarms are costly
Recall	How many sick patients were identified?	Missed cases are costly
F1	Single number balancing precision + recall	Sense of overall model quality relative to no model

EVALUATION METRICS

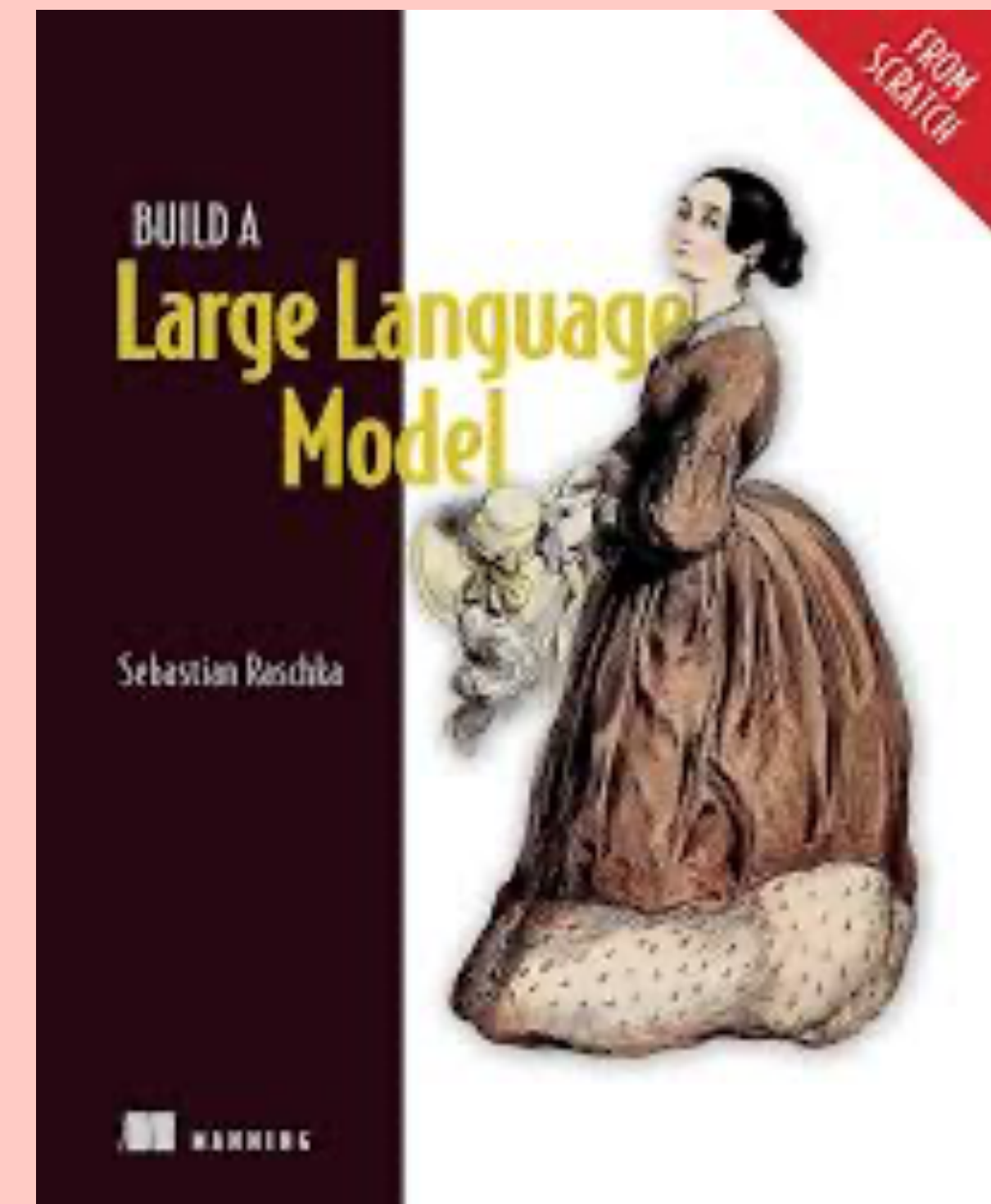
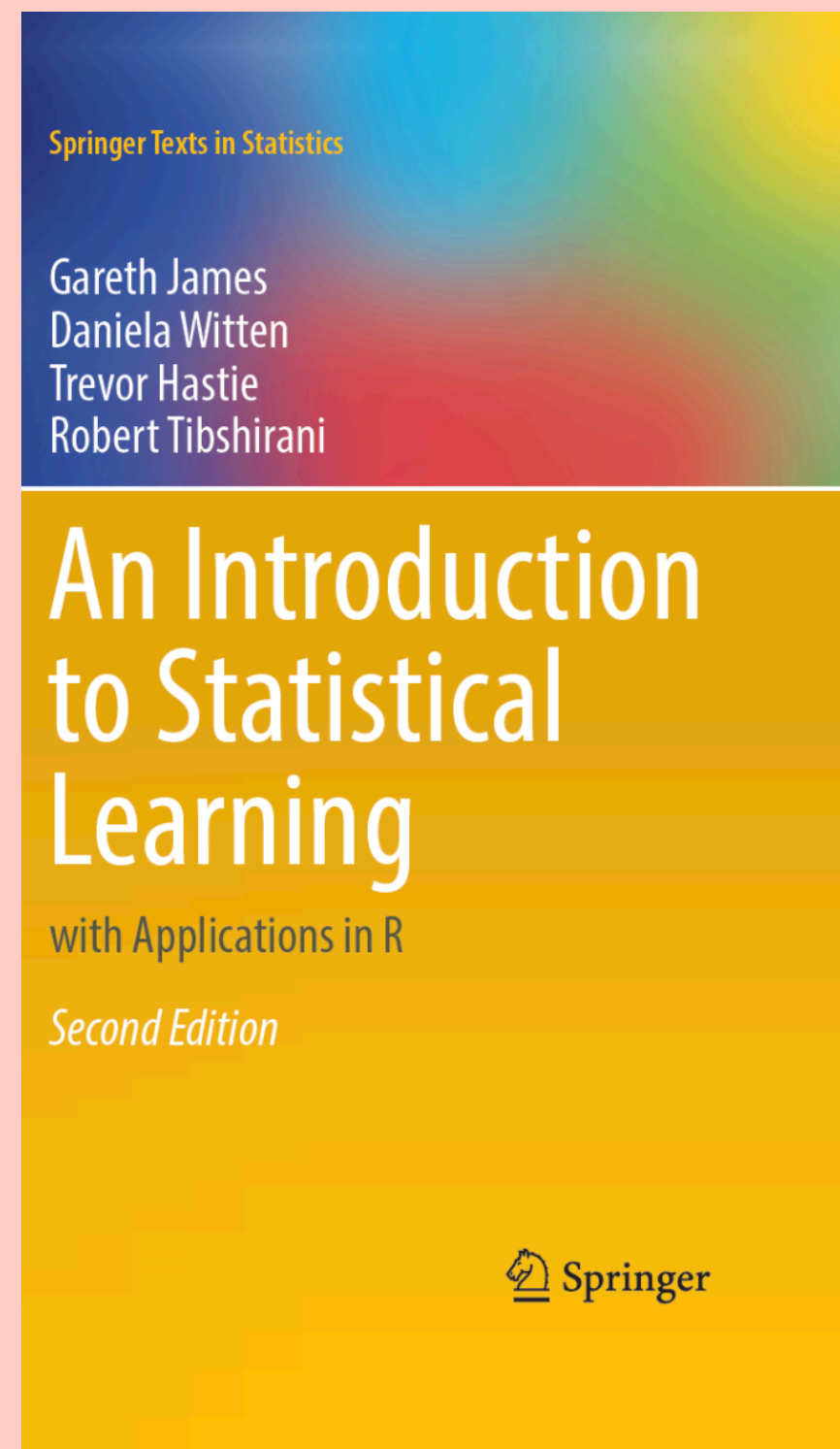
Metric	Description	Use Case
Mean Absolute Error (MAE)	Average of the absolute differences between predicted and actual values	All errors count equally
Root Mean Squared Error (RMSE)	Squares errors before averaging — penalizes large misses more heavily	Big mistakes are most costly
R^2	How much variation does the model explain	Sense of overall model quality relative to no model

THE TOOLKIT

STATISTICAL TOOLKIT

Tool	When to Use	Why?
Distribution	Data exploration, train/test split, monitoring	Answers what it “typical”
Bias/Variance	Underperforming model and need to understand why	Model too simple vs too sensitive
Cross-validation	Performance real or lucky	Looks at multiple splits of data
Regularization	Need to prevent/fix overfitting	Model fitting defense
Evaluation Metrics	Understanding model performance	Goes beyond predictive accuracy

FURTHER LEARNING





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